

Hong Kong Traffic Flow, Speed, and Junction Black site Distribution

GROUP ■ - GROUP REPORT

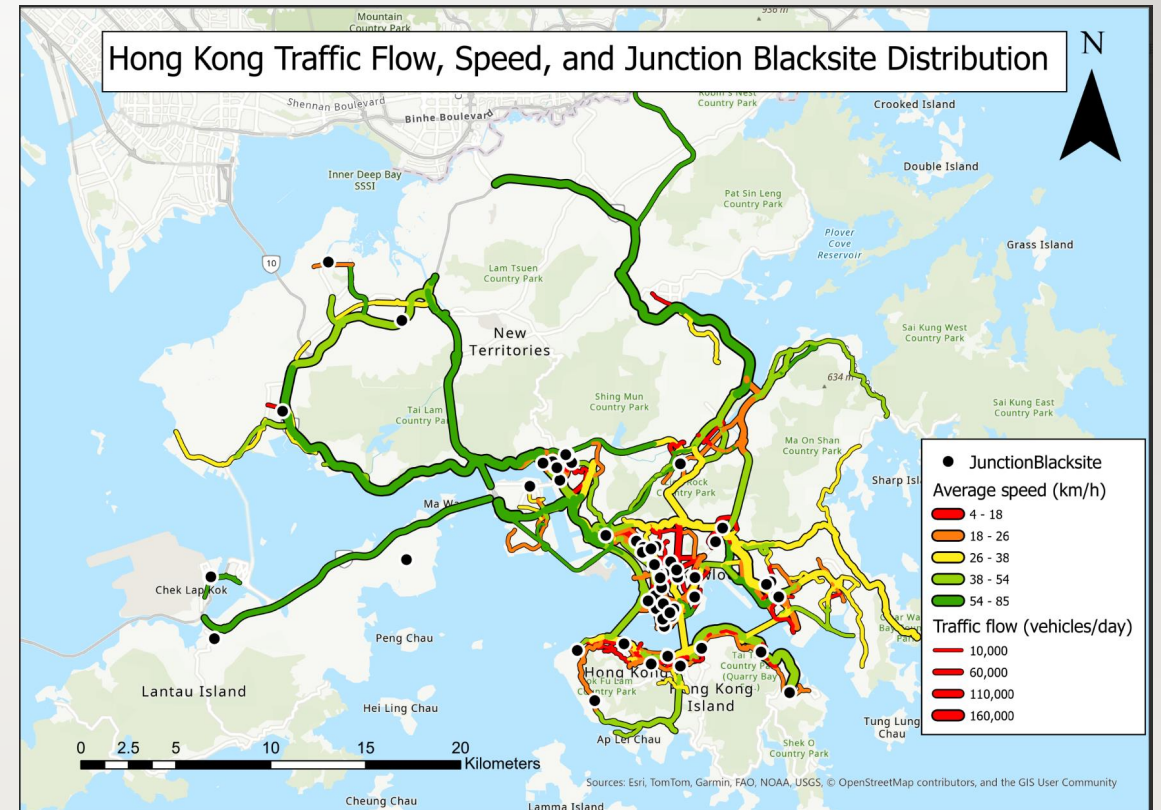




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Introduction

Cartographic principles to Hong Kong's road network

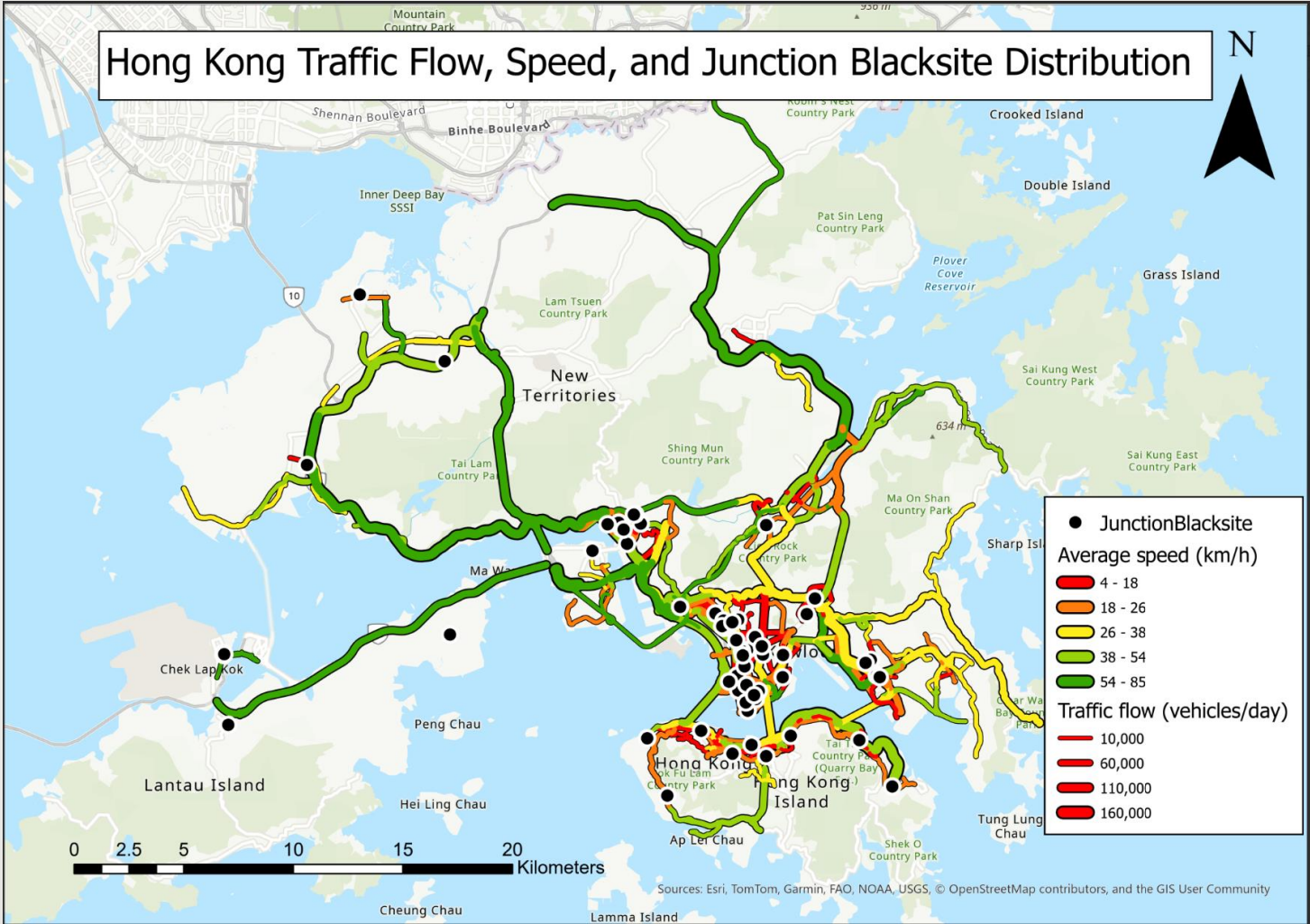
Visualizes **three** key elements:

- Traffic flow **volumes**
- Average vehicle **speeds**
- High-risk **junction blacksites**

The resulting map reveals critical spatial patterns to support

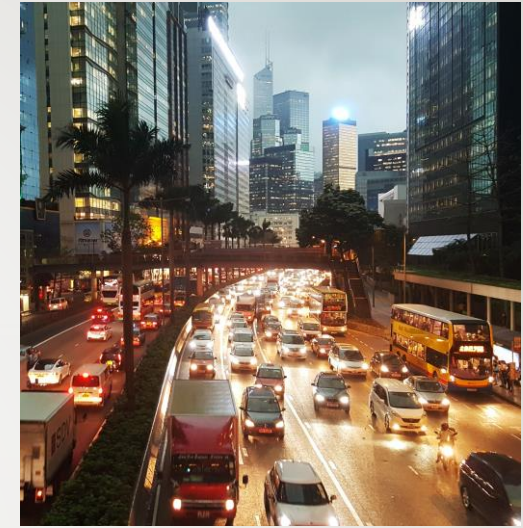
- Urban planning
- Traffic management
- Public safety decision

Our finalized map



Project Rationale: Traffic Congestion

- **High Average Congestion:** Hong Kong maintains a **47.7% average congestion rate**. Speeds during peak hours frequently drop **below 25 km/h**.
- **Heavy Volume:** In 2024, the average daily vehicle-kilometer reached **39.69 million**.
- -> high relevance to daily lives



Project Rationale: Safety Concerns

Junction Blacksites

- 6+ pedestrian accidents / 9+ annual casualty accidents
- -> map might reveal spatial correlation between **speed**, **volume**, and **accidents**



Project Rationale: Project importance

- **Importance:** The data indicates significant traffic slowdowns in specific regions, requiring spatial analysis.
- **Intended Audience:**
- Governmental departments (Transport Department, Highway Department) , Construction consultants and contractors
 - Inform **urban planning** and **traffic management**
 - Support targeted **safety interventions**
- Commuters for information on slowdowns, accidents, and safer road use



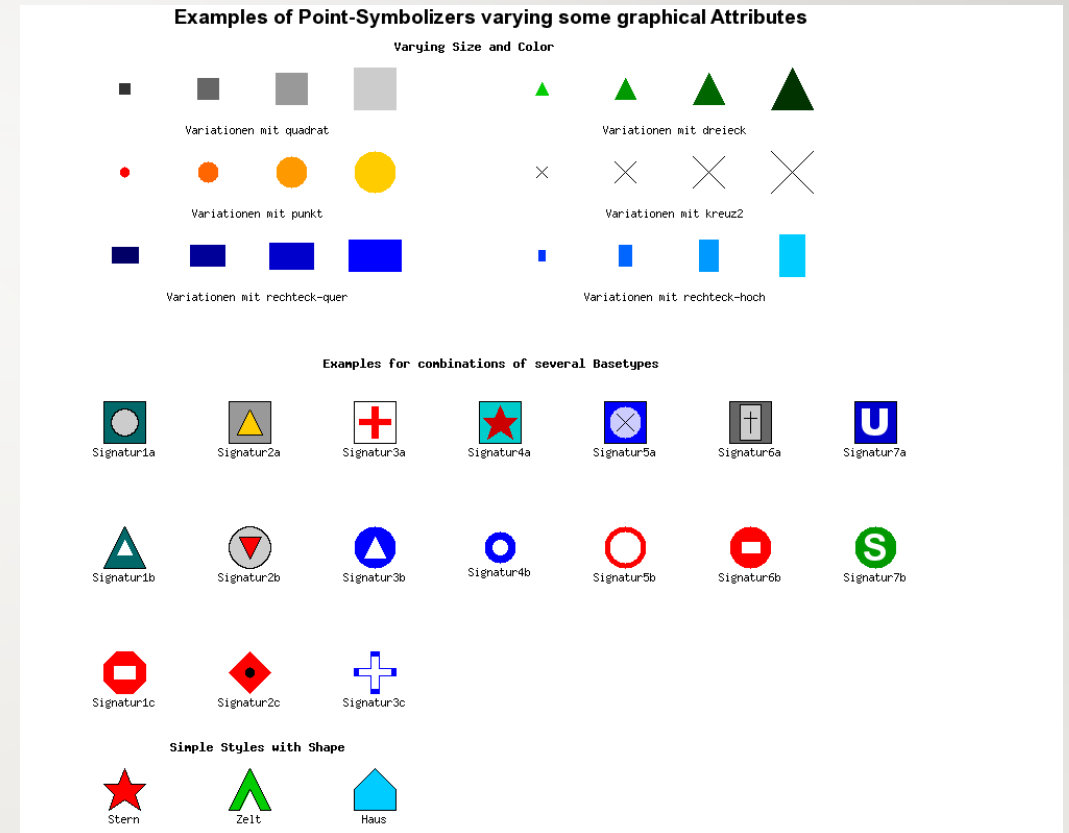
Project Objectives

Objectives:

- Apply cartographic principles in thematic map production
- Provide **insights** for urban planning and traffic management
 - Show **relationship** between traffic speed, flow, and accident locations
- Assist decision-making and future planning (**for urban planners**)
 - Improve **traffic safety** and **road efficiency**

Project Deliverables

- Use **symbology** with appropriate **visual variables** (size, hue)
 - respecting perception thresholds.
- Include all **essential map elements** (title, legend, scale bar, etc.)
- Evaluate map **effectiveness**
- Identify **strengths** and **limitations**



Map Theme

Map Title: Hong Kong Traffic Flow, Speed, and Junction Blacksite Distribution

Primary Theme: Flow Map

- **Bidirectional** roads – no arrows
- Line **width** reflect traffic flow volum
- **Hue** (**red** to **green**) = average speed

Average speed (km/h)

4 - 18

18 - 26

26 - 38

38 - 54

54 - 85

Traffic flow (vehicles/day)

10,000

60,000

110,000

160,000



Map Theme

Secondary Theme: Dot Map

- Sparse and discrete blacksite data
- Each dot = one high-risk junction
- Uniform, nominal data (belong to the same class, at different locations)

● JunctionBlacksite



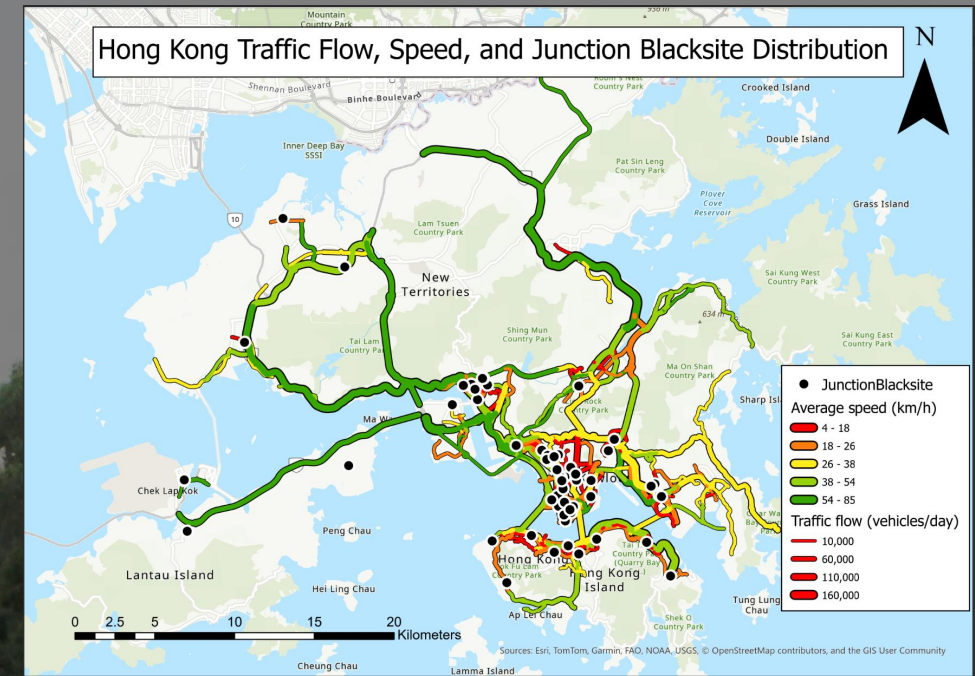
Geographic Coverage: whole Hong Kong

Show **highways** and **main roads** across **Hong Kong**

- Includes connections between New Territories, Kowloon, and Hong Kong Island

Some roads are **unavailable** as

- Missing traffic flow or speed data
- Not included in the shapefile
- E.g. San Tin Highway



Data unavailable



Data sources

Shape file for road networks	CSDI Portal (Lands Department, 2024)
Junction blacksite locations	
Traffic flow (vehicles/day)	Data.gov.hk (Government Logistics Department, 2024)
Traffic speed (km/min)	



- Official, recent data sources -> **reliable** and **relevant** data

Methodology – Data Cleaning

Traffic Speed data

$$\text{Speed} = \frac{\text{Length (km)} \times 60}{\text{Time (min)}}$$

1. Speed calculations (km/min) to (km/h)
2. Directional handling: lengths and times were averaged
3. Name standardization: converted to **uppercase** -> match the name in the shape file

Road/Street	Distance (km)		Journey Time (min)		Average d		Average tra
	A to B	B to A	A to B	B to A	(km)	(h)	
Aberdeen	1.60	1.70	1.90	1.90	1.65	0.031667	52.10526
Aberdeen	2.90	3.10	5.90	3.10	3	0.075	40
Airport R	0.70	0.00	1.00	0.00	0.7	0.016667	42
Albany R	0.20	0.00	3.50	0.00	0.2	0.058333	3.428571
Ap Lei Ch	1.90	1.30	3.10	1.90	1.6	0.041667	38.4
Argyle St	2.20	2.70	6.50	10.70	2.45	0.143333	17.09302
Austin Rd	1.00	1.00	5.20	3.40	1	0.071667	13.95349
Austin Rd	0.90	0.90	3.10	1.40	0.9	0.0375	24
Belcher's	0.00	1.00	0.00	5.10	1	0.085	11.76471
Berwick S	0.30	0.00	2.30	0.00	0.3	0.038333	7.826087
Bonham F	1.40	0.60	5.30	3.20	1	0.070833	14.11765
Boundary	2.50	0.00	8.40	0.00	2.5	0.14	17.85714
Bulkeley	0.20	0.00	1.00	0.00	0.2	0.016667	12
Bute Stre	0.10	0.00	0.20	0.00	0.1	0.003333	30
Butterfly	0.50	0.00	1.40	0.00	0.5	0.023333	21.42857
Cadogan	0.80	0.30	2.20	1.30	0.55	0.029167	18.85714
Caine Ro	0.90	0.00	4.30	0.00	0.9	0.071667	12.55814
Canal Ro	2.00	1.60	4.80	2.80	1.8	0.063333	28.42105
Canton R	0.70	1.20	2.90	4.60	0.95	0.0625	15.2
Castle Pe	0.70	2.30	0.90	8.40	1.5	0.0775	19.35484
Castle Pe	1.30	1.30	3.00	2.70	1.3	0.0475	27.36842
Castle Pe	4.80	4.90	7.40	7.20	4.85	0.121667	39.86301
Castle Pe	1.10	1.10	2.70	1.70	1.1	0.036667	30
Castle Pe	0.90	0.90	1.00	0.90	0.9	0.015833	56.84211
Castle Pe	1.90	2.00	4.90	3.80	1.95	0.0725	26.89655
Castle Pe	1.70	1.70	1.80	4.90	1.7	0.055833	30.44776
Castle Pe	2.40	2.40	5.00	5.30	2.4	0.085833	27.96117
Castle Pe	4.30	4.30	4.60	4.90	4.3	0.079167	54.31579
Castle Pe	2.20	2.20	6.70	3.00	2.2	0.080833	27.21649
Castle Pe	1.80	1.80	5.40	4.40	1.8	0.081667	22.04082
Castle Pe	5.20	5.60	11.20	12.30	5.4	0.195833	27.57447
Catchick	0.40	0.00	1.90	0.00	0.4	0.031667	12.63158
Carmarth	0.00	0.50	3.10	3.30	0.7	0.048333	14.48333

Methodology – Data Cleaning

- **Traffic Flow Data (AADT):**
- **Scripted Parsing:** Developed a custom **Python script** to parse raw XML files and extract road names and daily vehicle counts.
- **Data Conversion:** Exported the parsed data into **CSV format** for GIS compatibility.
- **Name Standardization (match with shape file):**
 - Convert into uppercase
 - Expand abbreviations (Rd -> ROAD, C -> CENTRAL, etc.)

RoadName	AADT
TOLO HIGHWAY	157910
CHEUNG TSING HIGH	154800
KWUN TONG ROAD	149710
ISLAND EASTERN CO	148100
PRINCE EDWARD RO	137580
TUEN MUN ROAD	137560
TSUEN WAN ROAD	135730
GLOUCESTER ROAD	133720
VICTORIA PARK ROA	133720
HARCOURT ROAD	125100
YUEN LONG HIGHWA	122600
TSING KWAI HIGHW	120310
HONG CHONG ROAC	119980
TAI PO ROAD - SHA	117480
FANLING HIGHWAY	114960
CHATHAM ROAD NC	114520
CHATHAM ROAD SO	113760

```
import xml.etree.ElementTree as ET
import csv
import re

XML_FILE = "ATC_TRAFFIC_DATA_2024_GISDB.xml"
CSV_FILE = "hong_kong_roads_aadt.csv"

direction_map = {
    "N": "North",
    "S": "South",
    "E": "East",
    "W": "West",
    "C": "Central",
}

def clean_road_name(name):
    # Data cleaning: replace Rd with Road, St with Street
    name = name.replace("Rd", "Road").replace("St", "Street")

    # Data cleaning: replace N, S, E, W with full names
    for abbr, full in direction_map.items():
        if name.endswith(abbr):
            name = name[:-len(abbr)] + full
            break

    return name.strip()

# Parse the XML
tree = ET.parse(XML_FILE)
root = tree.getroot()

# We'll collect data here
data = []

# Find all <PStnList> elements (they are direct children of <dataroot>)
for pstation in root.findall("PStnList"):
    road_name_elem = pstation.find("RoadName")
    cur_aadt_elem = pstation.find("CurAADT")

    if road_name_elem is not None and cur_aadt_elem is not None:
        original_name = road_name_elem.text.strip()
        cleaned_name = clean_road_name(original_name)

        aadt_text = cur_aadt_elem.text.strip()
        try:
            aadt = int(aadt_text.replace(",", ""))
        except ValueError:
            aadt = None

        data.append((cleaned_name, aadt))

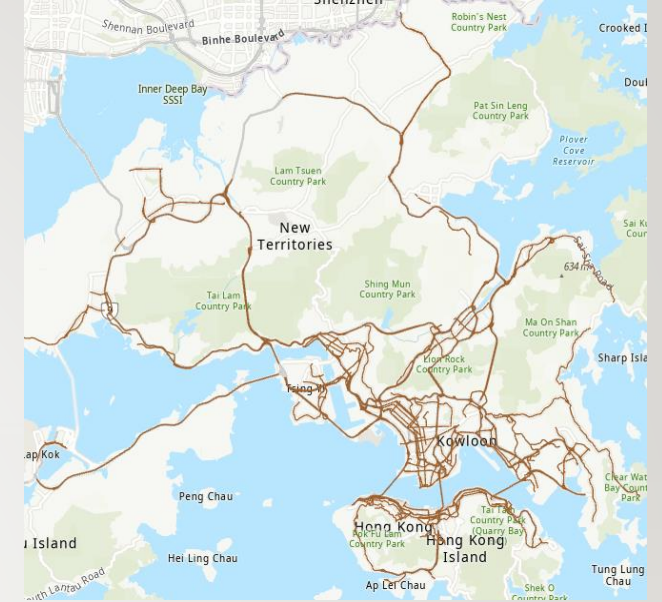
# Write to CSV
with open(CSV_FILE, "w", newline="", encoding="utf-8") as f:
    writer = csv.writer(f)
    # Header
    writer.writerow(["RoadName", "AADT"])
    # Data rows
    writer.writerows(data)

print(f"Processed {len(data)} roads.")
print(f"CSV file saved as: {CSV_FILE}")
```

Methodology – Data Filtering

Only keep the "most important" roads

- Definition: roads with high volume
- Threshold: 10000 vehicles/day
- Reduce visual clutter

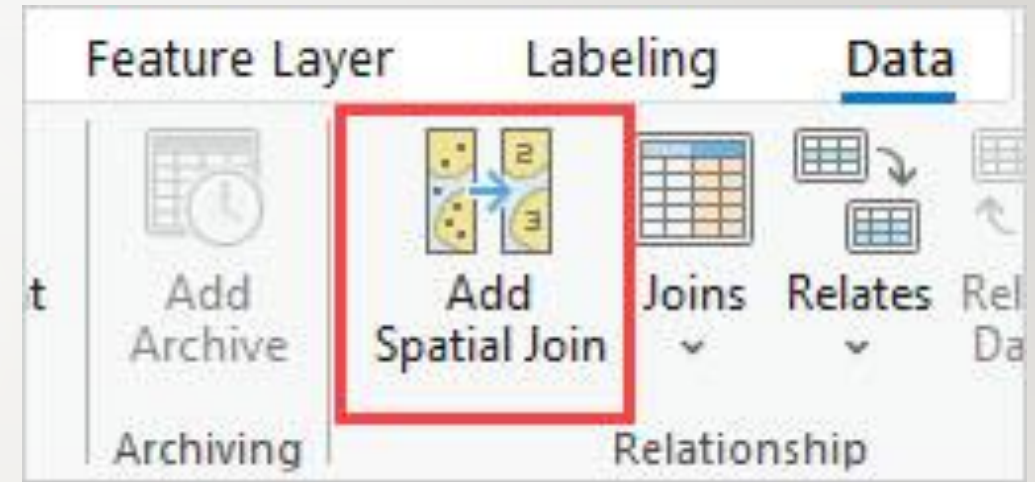


Feature	Geometry	Filter threshold
AADT	Line/Network	>10000 vehicles/day
Average Speed	Line/Network	None
Junction blacksites	Point	None

Methodology – ArcGIS Integration

GIS Integration (ArcGIS Pro)

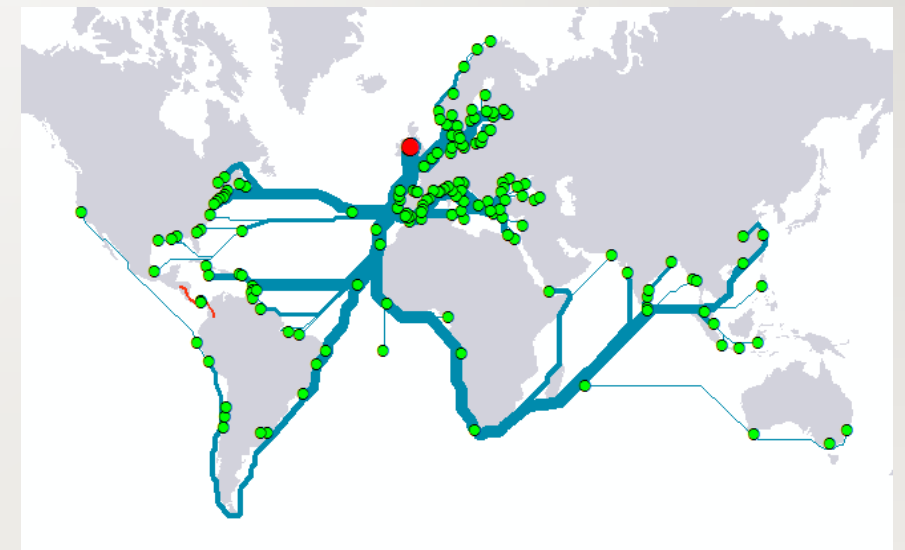
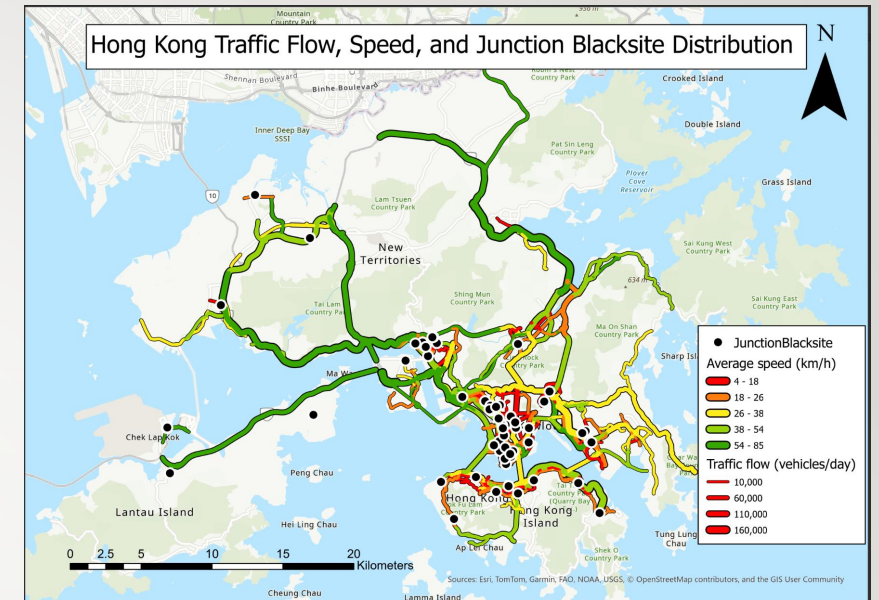
- **Spatial Join:** The processed Excel table was linked to the **road centerline shapefile**.
Tool: "Add Join".
- **Primary Key:** Joined via the “**Street English Name**” field.
- **Output:** Created a new “**Average Speed**” attribute within the spatial dataset.



Cartographic Design Decisions – Map Type

Map Type

- Hybrid thematic flow line map (primary)
- excellent for showing quantitative linear movement (e.g., traffic volume) along road networks
 - Uses road centerlines as the network base
 - **No arrows** are used to imply **bidirection**, and keep the map simple and **avoid visual clutter**



Cartographic Design Decisions – Secondary Map Type

Secondary Layer – **Junction Blacksites**

- Uniform point symbols
- Discrete, specific locations of junction blacksites
- Geometric instead of mimetic
 - Simplicity

● JunctionBlacksite

Primary symbology

Single Symbol

Symbol ●

Label

Description



Cartographic Design Decisions – Traffic Flow Data

- **Line width** represents **daily traffic flow** using four equal-interval classes: 3 px, 5 px, 7 px, and 9 px.
- Equal-interval classification was chosen for its simplicity and intuitive.
 - Difference between classes is 50000 vehicles per day

Traffic flow (vehicles/day)

— 10,000

— 60,000

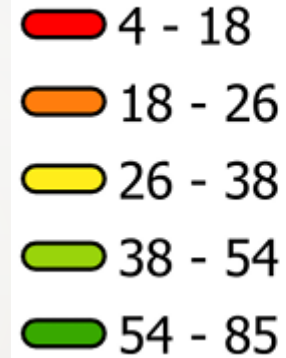
— 110,000

— 160,000

Cartographic Design Decisions – Traffic Speed Data

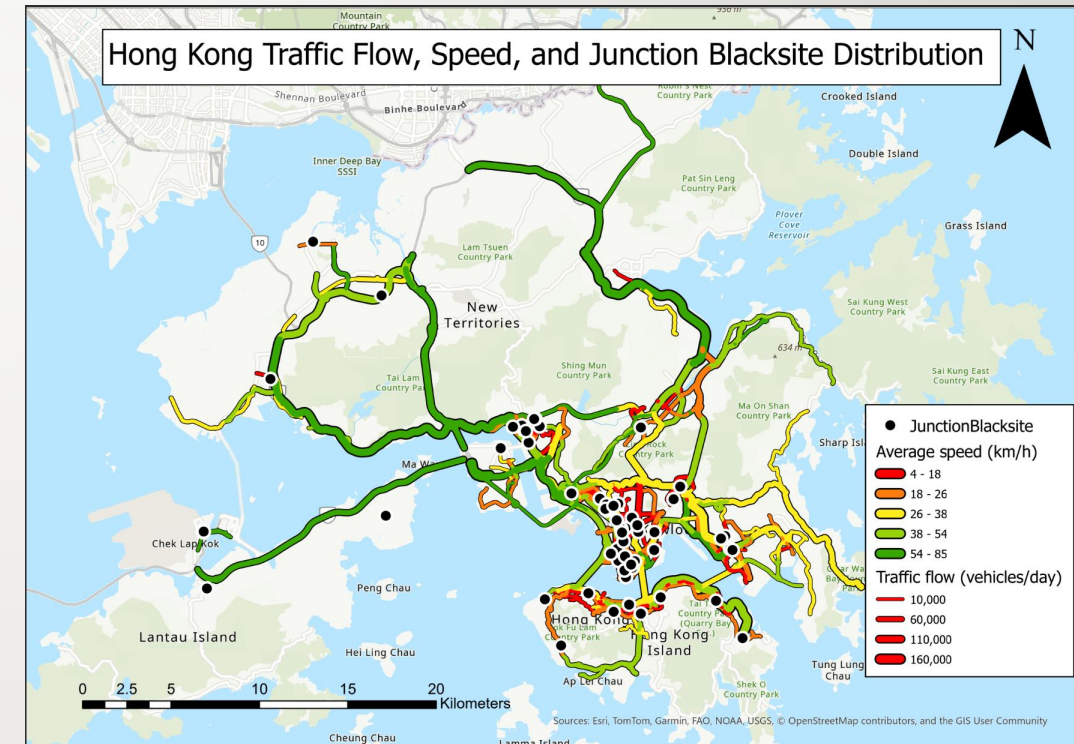
- Line colour shows vehicle speed using five **natural-breaks (Jenks) classes**, ranging from red (3–16 km/h) to green (54–84 km/h).
- 5 colour classes show sufficient **contrast**
- The red-green color scheme for traffic speed aligns with our **daily life expectation** and **map viewing habits** as on traffic update news.
- (Image source: *Morning Traffic Report* by TVB)

Average speed (km/h)



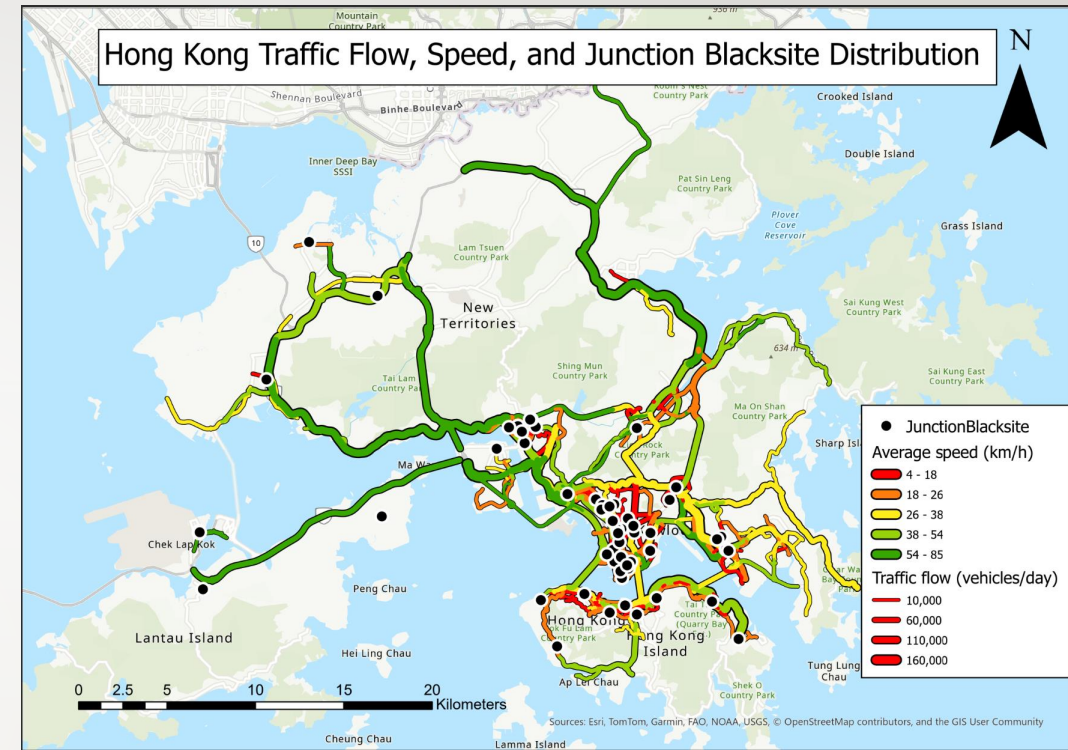
Cartographic Design Decisions – Junction Blacksite Data

- Junction blackspots are shown with **simple uniform black gray circles with white borders**.
- > Increase the **contrast** between the base map and the points
- All symbols exceed **minimum perception thresholds**, ensuring clear visibility and differentiation.



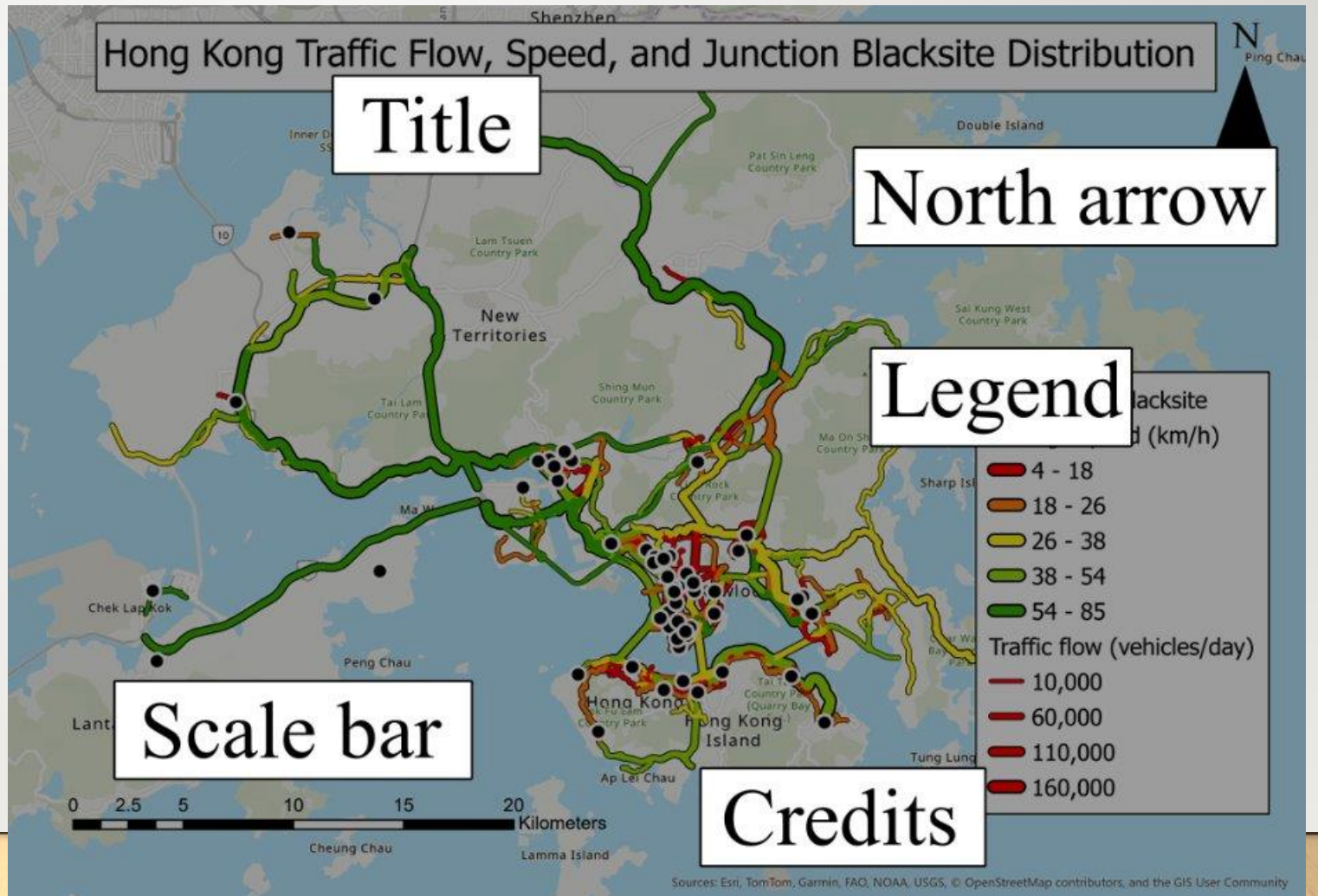
Primary Visual Variables

- Primary Visual Variables: **Hue** and **size** for **quantitative** data
- **Size** represents **flow magnitude** (ordered data)
- **Hue** represents speed intuitively (**Green** = **fast/safe**, **Red** = **slow/congested**)
- Follows best practices: Size for ratio/interval data, Hue for universal effectiveness
- Avoided secondary variables (e.g. patterns) to prevent visual clutter



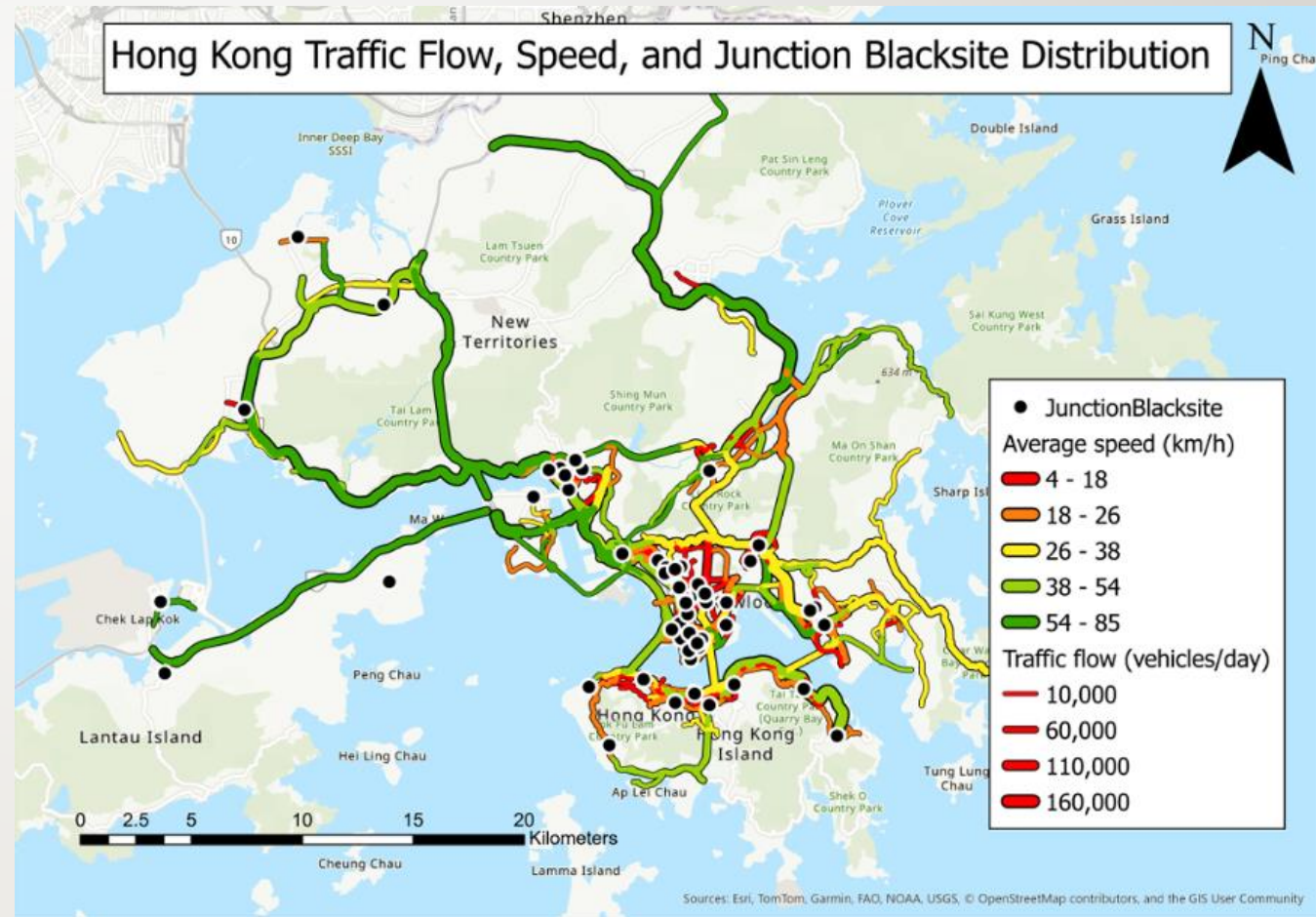
Essential Map Elements

- Essential Map Elements
- Core Map Elements:
 - Bold title
 - Comprehensive legend with full symbol explanations
 - Scale bar (in km)
 - North arrow
 - Complete data source credits



Layout Design and Visual Hierarchy

- Layout Design:
 - **Optical center** placement focused on Hong Kong
 - Clear figure-ground contrast (**dark** features on **light** base)
- Visual Hierarchy:
 - **Flow lines = most prominent**
 - Points = secondary
 - Background = minimal



Self-reflection - Strengths

- **Pattern Communication**
 - Shows higher correlation between blacksites and high-volume, low-speed roads
- **Content completeness**
 - Covers major roads with >10,000 vehicles/day using recent, reliable government data
- **Effective communication**
 - Overlays junction blacksites with traffic flow and speed to highlight congestion and safety priorities
- **Visual clarity**
 - Uses intuitive symbology (width, hue, points), balanced layout, and perceptual contrast for easy interpretation

Self-reflection – Limitations and Future Improvements

Limitations

- Potential visual clutter in ultra-dense urban areas
- Equal-interval flow classification slightly compress extreme values
- Static format limits temporal analysis

Future Improvements

- Dynamic animation for temporal traffic changes
- 3D relief to show terrain influence



Thank You For Listening

Q & A SECTION